Math 1351-013 Exam III-C March 26, 2004

Answer the problems on **separate** paper. You do <u>not</u> need to rewrite the problem statements on your answer sheets. Do your own work. **Show all relevant steps** which lead to your solutions. Attach this question sheet to the front of your answer sheets.

1. (40 pts) Find the derivative of each of the following explicitly defined functions (simplify where possible):

a. $f(x) = x(4x+2)^3$

b. $g(x) = \sqrt{\dfrac{x^2+4}{x^2+1}}$

c. $h(x) = \cos^{-1}(2x-3)$

d. $k(x) = \cos(\cos x)$

e. $j(x) = (2x-1)^x$

2. (16 pts) Find y' for each of the following implicitly defined functions:

a. $x^2 + 3xy - y^4 = 11$

b. $x^3 + y = x^2 - y^2$

3. (10 pts) Find the equation of the tangent line to the curve defined by $4x^2 + y^2 = 5xy - 5$ at the point (2,3).

4. (10 pts) A truck and car leave the intersection at the same. The car travels north at 80 kph while the truck heads east at 60 kph. How fast is the distance changing between the two vehicles 90 minutes later?

5. (10 pts) Use differentials to approximate $\sqrt[3]{27.2}$.

6. (10 pts) Using calculus, find the absolute maximum and absolute minimum values of $f(x) = 3x^5 - 8x^3 - 10$ on [-1,2].

7. (10 pts) Using calculus, find the absolute maximum and absolute minimum values of $f(x) = \sqrt{x}(4x - 3x^2)$ on [0,2].